Chapter Two

The Geology of Ethiopia and the Horn

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Objectives

At the end of this chapter, you should be able to:

- Explain the two major types of geologic processes.
- Analyze the geologic processes and the resultant land forms of Ethiopia and the Horn.
- Examine the formation of the Rift Valley.
- Recognize the current status of Ethiopian mineral endowment associated with geologic processes.

2.1. Introduction

- Geology is an **Earth science** that studies the;
  - ✓ **evolution of the earth**,
  - ✓ **the materials** of which it is made of, and
  - ✓ the **processes acting upon** them.
- Much of Geology is concerned with events that took place in the **remote past** and with features which are **far beneath the earth's surface** where no one can see them.
- A great deal of geological understanding must, therefore, be obtained by **inference** (assumption), using clues from what **can be seen** and **what can be measured**.

Intro……..

- Many clues not only the rocks and landforms which can be observed and studied at the Earth's surface, but also those provided by indirect methods such as

- **Geophysics** (e.g. studying earthquake waves which can penetrate deep beneath the earth's surface),
- **Geochronology** (methods for finding the ages of rocks, from the radioactive elements they contain).
- **Geochemistry** (analysis of the detailed composition of rocks which can give clues as to their origin)

Continental drift Theory

- Is a theory explain about the formation of continents.
- Australian Climatologist **Alfred Wegener** proposed the hypothesis that the continents were once **assembled together** as **a supercontinent**, called ***Pangaea***. It is a single huge continent.



Con't……

- Pangaea began to break up in **Mesozoic era**. It split into:
  - **Gondwanaland** in the south and
  - **Laurasia** in the north in Mesozoic era.



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- **Laurasia includes;** North America, Asia (except India) and Europe**.**



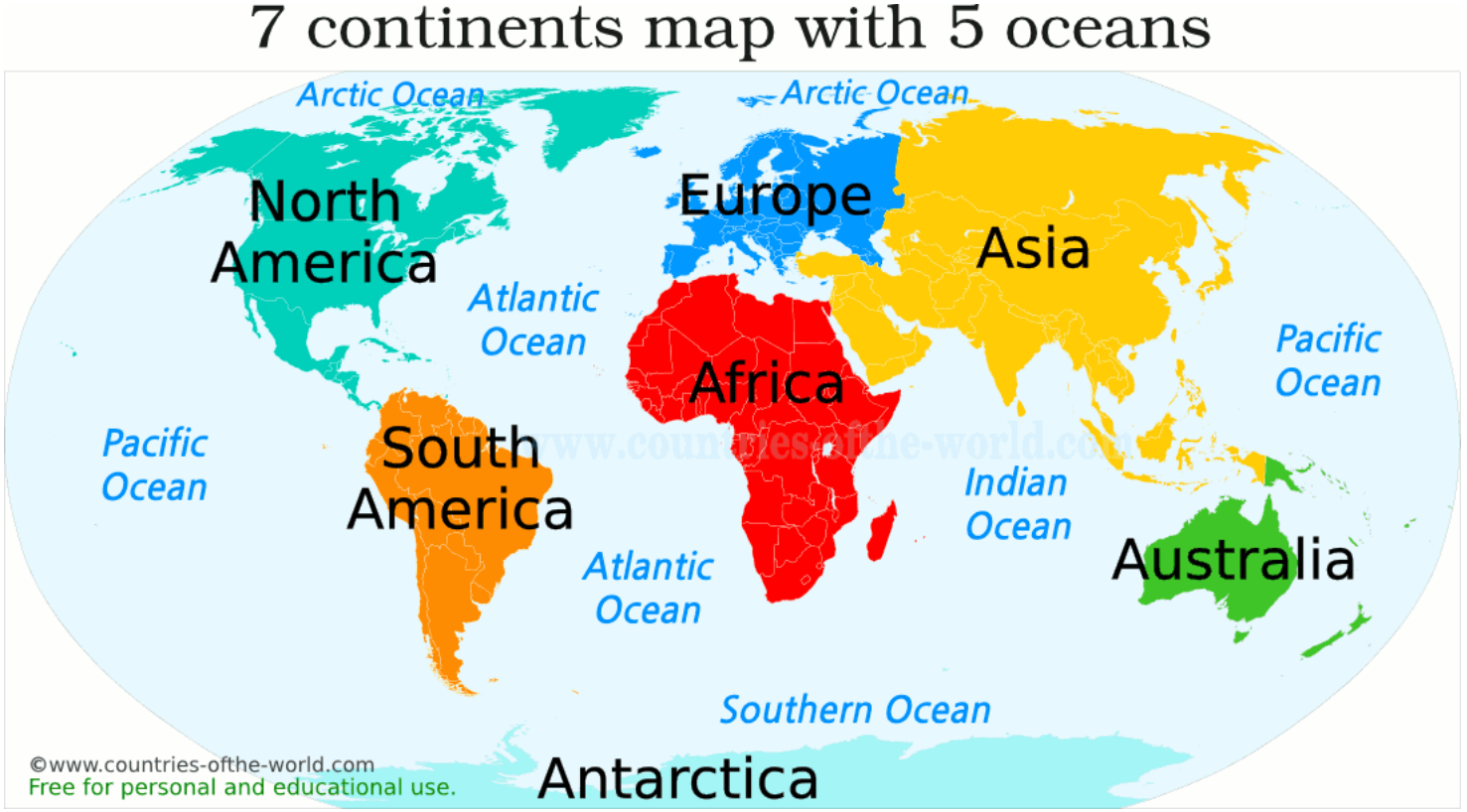
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- **Gondwanaland includes:** Africa, South America, Antarctica and Australia.



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- **Cenozoic era** - the present continents were formed.



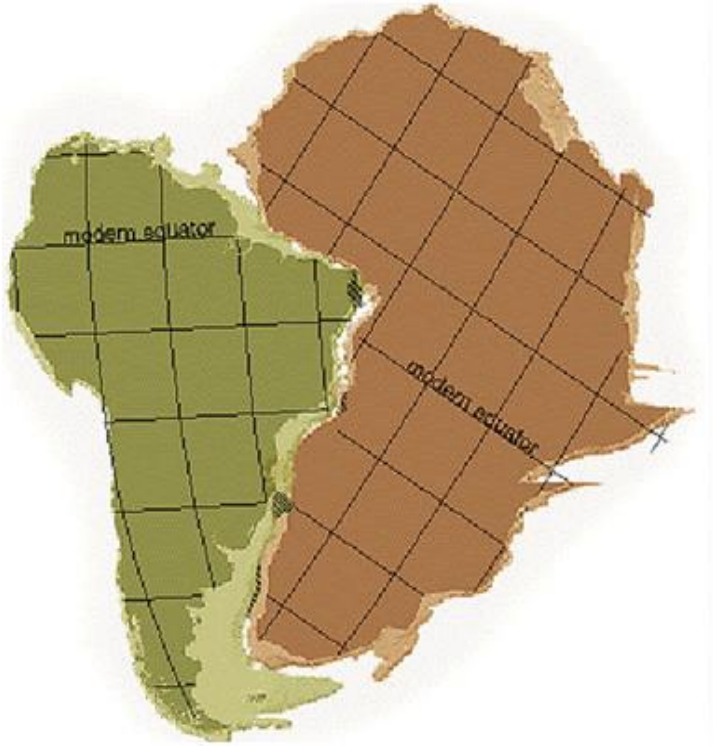
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Wegener's principal observations were:

1. ***Fit of the continents:*** The opposing coastlines of continents often fit together.
2. ***Match of mountain belts, rock types:*** If the continents are reassembled as Pangaea, mountains in West Africa, North America, Greenland, and Western Europe match up.
3. ***Distribution of fossils:*** The distribution of plants and animal fossils on separate continents forms definite linked patterns if the continents are reassembled.
4. ***Paleoclimates:*** rocks formed 200 million years ago in India, Australia, South America, and southern Africa all exhibited evidence of continental glaciations.

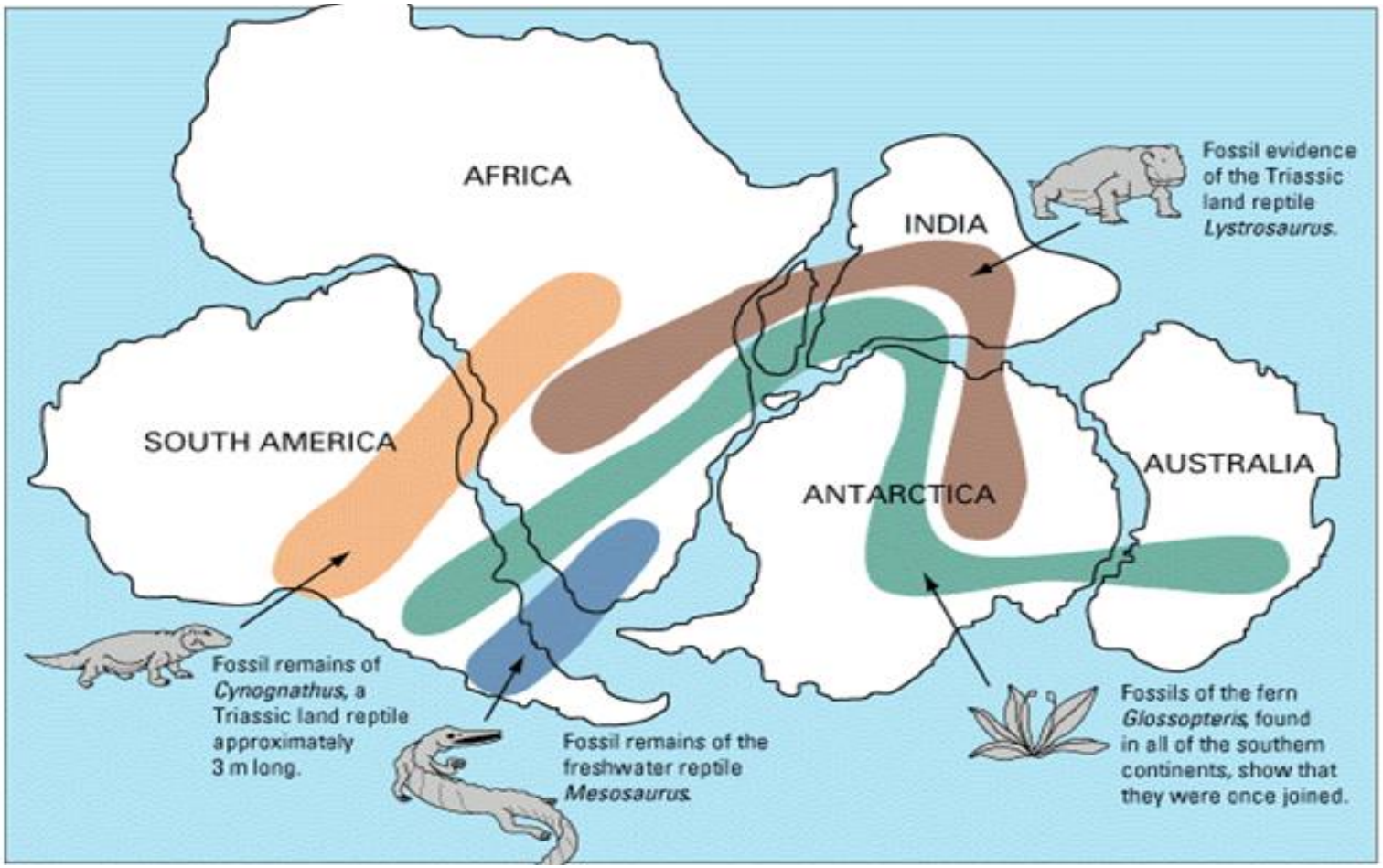
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Continental fit of Africa and the South America continents



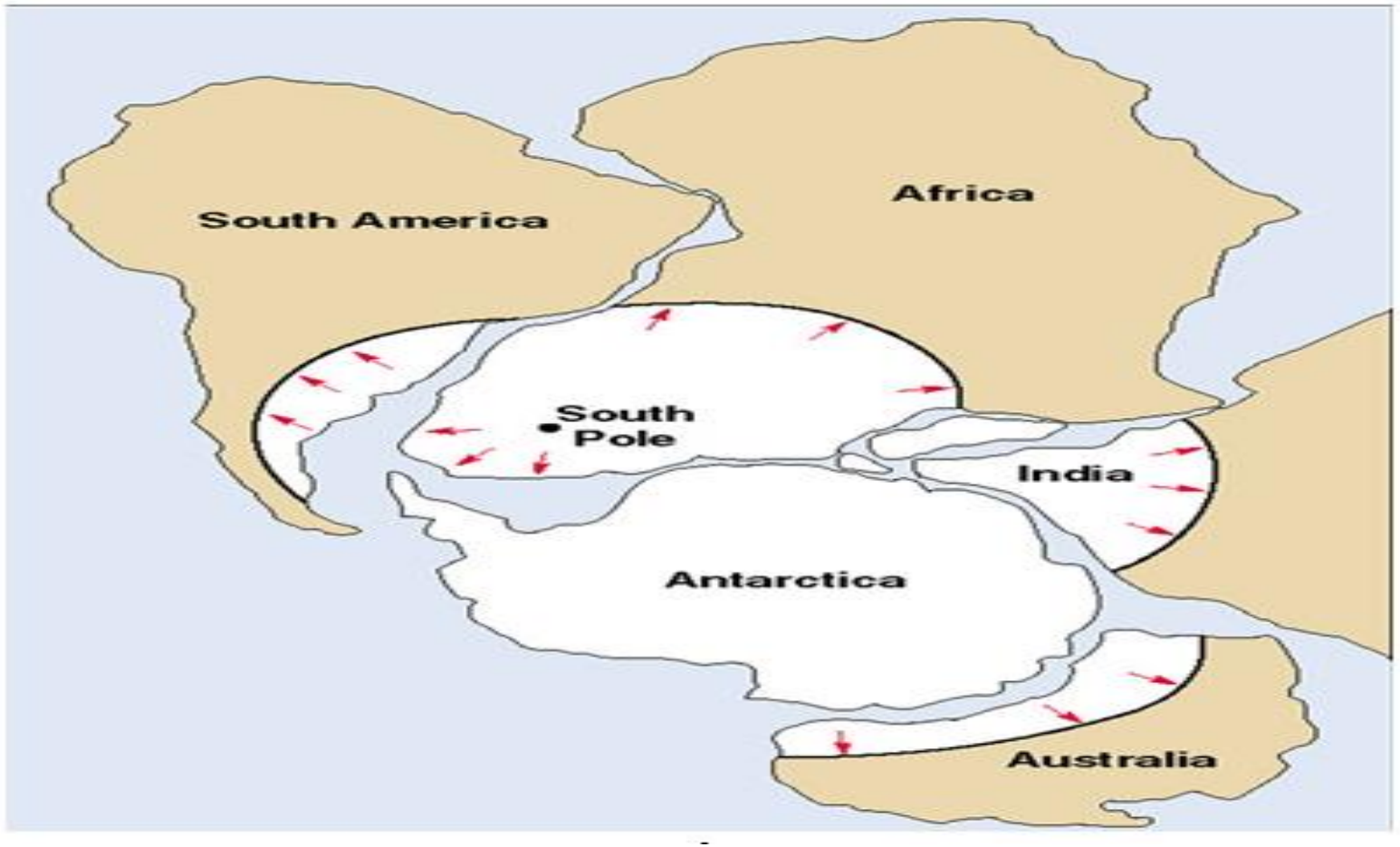
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The distribution of similar fossils



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Glaciated area, arrows indicate the direction of glacial movement



2.2. The Geologic Processes

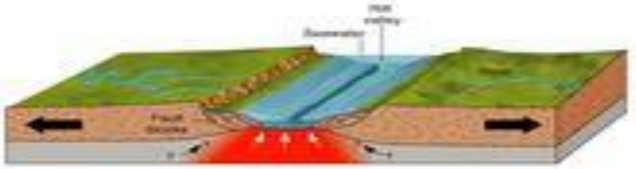
- Geology studies of how **Earth's materials, structures, processes** and **organisms** have changed over time.
- **Processes**: are **internal** or/and **external forces** that shape & reshape the earth within & without.
- They are of two broad types of Geologic processes;

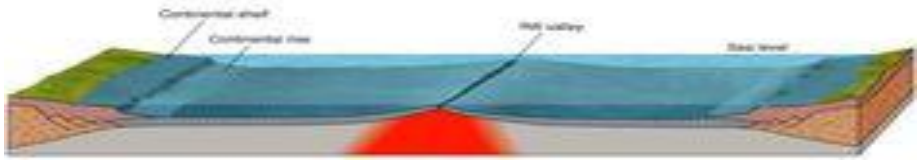


1. Indogenic forces (Internal): are **internal forces** which include **volcanic actions & tectonic forces.**

- Tectonic forces like (**folding, faulting, orogenesis** (mountain building), and **epeirogenesis** (slow rising and sinking of the landmass).
- responsible for creating **original landforms**







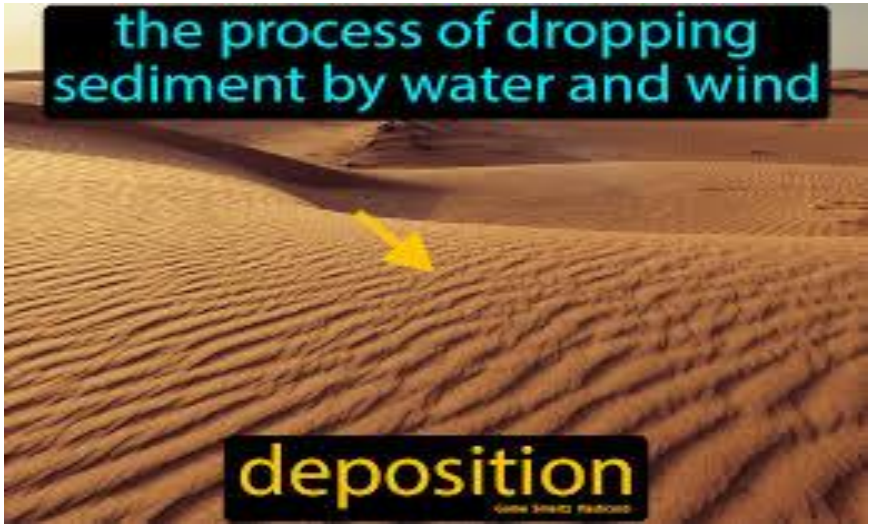
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2. Exogenic forces (External): are external forces functioning in the **outside section** of the earth's surface. They include

- **weathering** (breaking down of rocks),
- **mass transfer**,
- **erosion** and
- **deposition**.

- Responsible for **modifications** of original landforms.





- Generally, **any place's** present landforms are the result of the **joint actions of both internal & external forces**.

2.3. The Geological Time Scale and Age Dating Techniques

2.3.1. Geological Time Scale

- Is a scale used to measure the age of the earth.
- The geological time scale measures time on a scale involving four main units:
  - ✓ An ***epoch*** is the smallest unit of time on the scale and encompasses a period of millions of years.
  - ✓ Chronologically, epochs are clumped together into larger units called ***periods.***
  - ✓ Periods are combined to make subdivisions called ***Eras.***
  - ✓ An ***eon*** is the largest period of geological time.
- The four geological eras are commonly used to measure the age of the earth.

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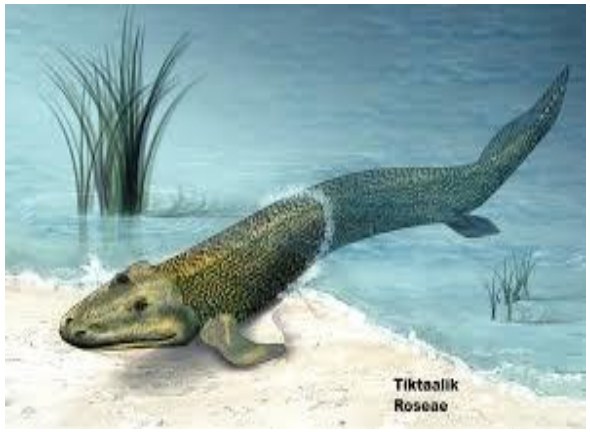
From the **oldest** to the **youngest** the sequence of the four eras is given in the table below.

S. No.	Geological Era	Time Range	% of GTS
1.	Precambrian	4.6 billion to 600 million years ago	87
2.	Paleozoic	600 million to 225 million years ago	8
3.	Mesozoic	225 million to 70 million years ago	3.4
4.	Cenozoic	70 million to the present	1.6

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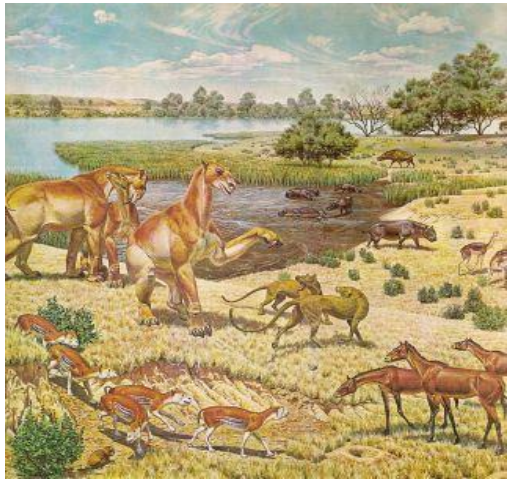
The Eras are given names that indicate the **kind of life** that existed in them.





- Before life appeared- Precambrian Era.
- Paleozoic Era - the age of invertebrates |





- Mesozoic Era - the age of reptiles while
- Cenozoic Era - the age of mammals.

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- These geological time divisions basically differ from each other in such characteristics as;
  - ✓ the relative **position of land** and **sea**,
  - ✓ the kind of **climate** and **most important** the **kind of animal** and **plant** life that developed and existed during that Era or period.
- The Earth is believed to have been formed approximately **4.6 billion years ago** and the earliest forms of life were thought to have originated approximately **3.5 billion years ago**.

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2.3.2. Age Dating Techniques

- Is a technique used to measure the age of rocks or fossils.
- There are two techniques of knowing the age of rocks.

A. Relative Dating

- Relative dating uses **geological evidence** to assign comparative ages of fossils.
  - ✓ Hence, we can use two ways to know the relative age of a rock:
  - ✓ One way is to **look at any fossils** the rock may contain.
  - ✓ The second way is to use the "**What is on top of the older rocks**?"

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B. Absolute Dating

- Also called numerical dating.
- Used to give rocks an actual date, or date range in number of years.
- This technique was developed with discovery of **radioactivity** in 1896. also known as Radiometric techniques.

Two of the major techniques include:

A. *Carbon-14 Technique:* organic matters

B. *Potassium-Argon Technique:* Inorganic matters

2.4. Geological Processes and the Resulting Landforms of Ethiopia and the Horn

- Let us see the geological processes and resulting land forms in Ethiopia based on the **four geological eras**.

2.4.1. The Precambrian Era Geologic Processes (4.5 billion - 600 million years ago)

- The **oldest** and **longest** geological era.
- Limited knowledge
- Despite prevailing uncertainties, **major geologic event** of the Precambrian Era was ***Orogenesis***
- As a result, the land was subjected to **intense folding**
- These processes resulted in the **formation of very huge mountains** & the **oldest rocks of the earth** known as **Basement Complex** or **Old Crystalline rocks** which are economically useful as they contain **metallic minerals**.

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The Precambrian rocks are overlaid by recent rock formations.

- However, as surface rocks covering **25%** of the land mass of the country; they are found exposed in the following areas:
- In the northern part: Western lowlands, parts of **northern** and **central** Tigray.
- **In the western Part**: Gambella, Benishangul-Gumuz (Metekel and Asossa), western Gojjam, western Wellega, Illuababora, and Abay gorge.
- **In the southern** Part: Guji, southern Omo, and parts of southern Bale and Borena.
- **In the eastern part**: Eastern Hararghe.

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2.4.2. The Paleozoic Era Geologic Processes (600million - 225 million years ago)

- The Paleozoic Era lasted for about **375 million** years.
- There were two major geological activities (external)

1. **Denudation** that refers to the **process of degradation** which crumbled the great mountain chains and
2. **Penplanation** which resulted in the formation of the **flat plateaus** & **plains** which represent the **present-day relief structure** of Ethiopia.

- Geologists believe that, this era left a gap in rock formation in Ethiopia
- Hence, no **major rock formation** took place during this era in the country.

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2.4.3. The Mesozoic Era Geologic Processes (225-70 million years ago)

- Mesozoic Era lasted for about ***155 million*** years.
- Markedly characterized by **internal process** followed by series of external activities.
- This process is called **Epeirogenesis**.
- Epeirogenesis is a vertical earth movement which refers to an **alternate slow sinking & rising** of the land.
- This process affected the whole present-day **Horn of Africa** and **Arabian landmass**.
- At the same time the land was tilted **eastward** and therefore **lower in the southeast** and **higher in the northwest**.
- Mesozoic era has three periods. **Triassic**, **Jurassic** and **Cretaceous**

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i). Triassic period of Mesozoic Era

- Marked the **sinking of the land** from southeast to northwest.
- Followed by **transgression of the sea** from the southeast (Indian ocean)
- As the land sank slowly the **sea invaded** it starting from Somalia and Ogaden and slowly spreading **northwestward**.
- This resulted in the deposition of **the first sedimentary rock** in Ethiopia as well as the region as a whole
- This deposit is called **Adigrat** s**andstone** (a place name found somewhere in Tigrai, Ethiopia)
- It is also known as **Lower sandstone**.
- The Adigrat sandstone is older and thicker in the southeast and progressively decreases in age and thickness northwest ward.

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ii). Jurassic period of Mesozoic Era

- Continued **sinking of the land** & **subsequent sea deepening**.
- This was followed by the deposition of the **second type of sedimentary layer** called **Hintalo/Antalo limestone**.

iii). Cretaceous period of Mesozoic Era

- Marked the **rising of the land** from northwest to the southeast.
- The rising of the land was followed by **regression of the sea** to its original position (Indian Ocean).
- In turn the retreating sea caused the deposition of **the third main type of sedimentary rock** in Ethiopia, called Upper sandstone.

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- The Upper sandstone, on the other hand, is thicker and younger (Upper Cretaceous) in the Southeast, while in the Northwest it is older and thinner.
- In conclusion, in Ethiopia at the end of Mesozoic era three different lithological stages of sedimentary layers witnessed namely; **Adigrat sandstone**, **Hintalo limestone** & **Upper sandstone**.
- These three layers are commonly called **Old Marine Sediments**.
- Hence, Mesozoic rocks are considered to have the **greatest potential for oil** and **gas deposits**.

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2.4.4. The Cenozoic Era Geologic Processes (70million years ago - Present)

- The preset topography of Ethiopia was largely emerged during the **Cenozoic Era**, especially due to geological processes of **Tertiary Period**.
- The major geological activities include;
  1. The uplifting of the land which continued from the previous era.
  2. The formation of the Great Rift Valley
     - ✓ Covers 7, 200 km, 5600 km in Africa, 1700 in Ethiopia and Eritrea

3. **Quaternary volcanic eruption & deposition.**

- During the Pleistocene, volcanic activity continued, especially in the Rift Valley.
- Features such as Erta Ale and Dubbi volcanoes were formed, along with lava ridges and hot springs